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CMPE 126
Programming Exam 3
Search.pdf

I will be hashing the students into the array rather than adding them in one by one. It will be based on the kind of hashing for cache where a subset of the student's ID will be the index. This is determined by the size of the array being 64, so optimally the number of digits will be $\log_2(64) = 6$ digits meaning the last 6 digits of the student's id will be used as the index. However, our array only holds indexes with two digits so we will go with the last two instead. To account for collisions, we will be implementing a linear probing hash function.

Our addStudent & addStudents function were not too difficult as we implemented a hashing function. However had we appended each student normally into the array, I would have sorted the array and implemented binary search for the addStudent & search. For the addStudents, I would have sorted that as well and merged this array with the original. Of course in my case, all I have to do was hash each student again so it was all $O(1)$.

Sources:

<https://stackoverflow.com/questions/7663709/how-can-i-convert-a-stdstring-to-int>
<https://stackoverflow.com/questions/7549874/how-do-i-fix-an-ambiguous-function-call>